M. Rohaan 221-2327 Quiz# Sec. E

Q#01:-

a) $P(X>5)=\frac{1}{3}$, $a=2$, $b=?$

$P(X>5)=\frac{b-5}{b-2}$ ∵ $X\sim U(2,b) \Rightarrow b-2$

$X>5 \Rightarrow b-5$

$\frac{b-5}{b-2}=\frac{1}{3} \Rightarrow (b-5)3=1(b-2)$

$2b=13 \Rightarrow \boxed{b=\frac{13}{2}}$

b) ∵ $X\sim U(10,20)$, $P(12\le X\le 17)$

↳ $20-10=10$

$17-12=5$

$P(12\le X\le 17)=\frac{5}{10}=\frac{1}{2}$

c) $M_X(t)=\frac{e^{tb}-e^{ta}}{t(b-a)}$, $t\neq 0$

$M_X(t)=E(e^{tX})$

$M_X(t)=\int_a^b e^{tx} f_X(x)\,dx$

$X\sim U(a,b)$, $f_X(x)=\frac{1}{b-a}$

$M_X(t)=\int_a^b e^{tx}\frac{1}{b-a}\,dx$

$$M_X(t) = \frac{1}{b-a}\int_a^b e^{tx}\,dx$$

$$\int e^{tx}\,dx = \frac{e^{tx}}{t}, \quad (t \neq 0$$

So, $$M_X(t) = \frac{1}{b-a}\left[\frac{e^{tx}}{t}\right]_a^b$$

$$M_X(t) = \frac{1}{b-a}\left(\frac{e^{tb} - e^{ta}}{t}\right)$$

$$M_X(t) = \frac{e^{tb} - e^{ta}}{t(b-a)}$$

Q#02:-

a) $P(X>2) = 0.1353$; $\lambda = ?$

$$P(X>2) = e^{-2\lambda}$$

$$e^{-2\lambda} = 0.1353$$
$$-2\lambda = \ln(0.1353)$$
$$\lambda \approx 1$$

b) $P(X>8)$

$$\frac{1}{\lambda} = 5 \Rightarrow \boxed{\lambda = \frac{1}{5}}$$

$$P(X>8) = e^{-\lambda \cdot 8} = e^{-8/5} \approx 0.2019$$
$$P(X>8) = e^{-1.6} \approx 0.2019$$

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c) $P(X>s+t \mid X>s) = P(X>t)$

$$= \frac{P(X>s+t \text{ and } X>s)}{P(X>s)}$$

$\because P(X>s+t \text{ and } X>s) = P(X>s+t)$

$$P(X>s+t \mid X>s) = \frac{P(X>s+t)}{P(X>s)}$$

$P(X>s+t) = e^{-\lambda(s+t)}$ and $P(X>s) = e^{-\lambda s}$

$$P(X>s+t \mid X>s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}$$

$P(X>s+t \mid X>s) = P(X>t)$

Q#03:-

a) $Exp(\lambda) = Gamma(1,\lambda)$

$X_1 \sim Gamma(1,\lambda)$, $X_2 \sim Gamma(1,\lambda)$

$X_1 + X_2 \sim Gamma(1+1,\lambda) = Gamma(2,\lambda)$

b) $E[X] = \frac{\alpha}{\lambda} = \frac{3}{2}$

$V(X) = \frac{\alpha}{\lambda^2} = \frac{3}{2^2} = \frac{3}{4}$

c) $M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^{\alpha}$, $t<\lambda$

$X \sim Gamma(\alpha,\lambda)$

$$f_X(x) = \frac{\lambda^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x>0$$

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The MGF is ; $M_X(t) = E(e^{tx}) = \int_0^\infty e^{tx} f_X(x)\,dx$

$$M_X(t) = \int_0^\infty e^{tx} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}\,dx$$

$$M_X(t) = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\lambda-t)x}\,dx$$

$$\int_0^\infty x^{\alpha-1} e^{-cx}\,dx = \frac{\Gamma(\alpha)}{c^\alpha}, \quad c>0$$

$$c = \lambda - t$$

$$\lambda - t > 0 \Rightarrow t < \lambda$$

$$\int_0^\infty x^{\alpha-1} e^{-(\lambda-t)x}\,dx = \frac{\Gamma(\alpha)}{(\lambda-t)^\alpha}$$

$$M_X(t) = \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha)}{(\lambda-t)^\alpha} = \frac{\lambda^\alpha}{(\lambda-t)^\alpha}, \quad t < \lambda$$

Q#04:-

a) When $\alpha = \beta$, both parameters are equal, so the distribution is symmetric about $x = \frac{1}{2}$

b) $X \sim Beta(2,3)$, find $E[X]$

$$E[X] = \frac{\alpha}{\alpha+\beta} = \frac{2}{2+3} = \frac{2}{5}$$

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c) $E[X] = \frac{\alpha}{\alpha+\beta}$

$X \sim Beta(\alpha, \beta)$, $f(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}$

$, 0 < x < 1$

$E[X] = \int_0^1 x \cdot f(x)\, dx$

$E[X] = \int_0^1 x \cdot \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}\, dx$

$= \frac{1}{B(\alpha,\beta)} \int_0^1 x^{\alpha}(1-x)^{\beta-1}\, dx$

$E(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx$

$\int_0^1 x^{\alpha}(1-x)^{\beta-1}\, dx = B(\alpha+1, \beta)$

$E[X] = \frac{B(\alpha+1,\beta)}{B(\alpha,\beta)}$

$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$

$B(\alpha+1,\beta) = \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}$

$E[X] = \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$

$$E[X] = \frac{\Gamma(\alpha+1)\,\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\alpha+\beta+1)} \qquad \because \Gamma(\alpha+1) = \alpha\,\Gamma(\alpha)$$

$$\because \Gamma(\alpha+\beta+1) = (\alpha+\beta)\,\Gamma(\alpha+\beta)$$

$$E[X] = \alpha \frac{\Gamma(\alpha)\,\Gamma(\alpha+\beta)}{\Gamma(\alpha)(\alpha+\beta)\,\Gamma(\alpha+\beta)} = \frac{\alpha}{\alpha+\beta}$$